

Monte Carlo Proxy Pricing

A step-by-step methodology for European, American, Asian, barrier, cliquet, SLV cliquet, and three-underlying basket cliquet instruments

This guide derives the transformations, estimators, regression bases, interpolants, variance-reduction methods, dynamic-programming equations, and validation metrics used in the proxy-pricing experiments.

Instrument	Effective state	Current preferred fitter
European call	1D d1-like moneyness	PCHIP default; Bernstein optimized
American put	Spot and exercise time	PCHIP continuation recursion
Arithmetic Asian	1D adjusted moneyness after reduction	PCHIP default; Akima optimized
Barrier call	Spot plus alive/hit flag	PCHIP + Brownian bridge
GBM cliquet	1D accrued clipped return	Bounded Chebyshev degree 19
Single-name SLV cliquet	Accrued, spot, variance	Local/spectral hybrid
3-asset SLV basket cliquet	Accrued, 3 spots, 3 variances	Sparse baseline; enrichment needed

Research implementation: [proxy_pricing repository](#)

1. Scope and guiding principle

A pricing proxy is a fast approximation $V_{\text{hat}}(x,t)$ to an expensive risk-neutral value $V(x,t)$. The central difficulty is not regression alone. It is choosing a Markov state, transforming the payoff geometry, generating unbiased low-noise labels, fitting without violating known bounds, and testing in the tails.

The experiments support one broad rule: use product structure to reduce dimension first, then choose the simplest smoother compatible with the remaining geometry. A one-dimensional curve and a seven-dimensional SLV surface should not be forced into the same numerical representation.

Contents

Part	Topic
I	Risk-neutral valuation, Monte Carlo labels, and variance reduction
II	Target transforms, Chebyshev/Bernstein bases, PCHIP, Akima, Bezier
III	European and Asian one-dimensional reductions
IV	American optimal stopping and barrier options
V	Single-name cliquet and SLV extensions
VI	Three-underlying SLV basket cliquet
VII	Generic high-dimensional workflow and validation checklist

What the reported error means

Unless stated otherwise, relative error is $\text{abs}(\text{proxy} - \text{benchmark})$ divided by $\max(\text{abs}(\text{benchmark}), 0.01)$. The 0.01 floor prevents an economically tiny absolute discrepancy near zero from dominating a percentage statistic. Signed and absolute errors are retained separately.

2. Risk-neutral valuation foundation

2.1 Conditional expectation

Under a risk-neutral measure Q , with short rate r and state X_t , a European-style discounted payoff H has value

$$V(t, x) = E_Q[\exp(-\int_t^T r_u du) H(X_T) \mid X_t = x].$$

Path-dependent contracts become Markov after augmenting X_t with sufficient path statistics: a running sum for an Asian, accrued clipped return for a cliquet, and current variances for SLV.

2.2 GBM transition

For constant parameters under Q :

$$dS_t / S_t = (r - q) dt + \sigma dW_t.$$

Integrating $d \log(S_t)$ gives

$$\log(S_T/S_t) = (r - q - \sigma^2/2) \tau + \sigma \sqrt{\tau} Z, \\ Z \sim N(0, 1).$$

Therefore

$$S_T = S_t \exp((r - q - \sigma^2/2) \tau + \sigma \sqrt{\tau} Z).$$

This exact transition removes time-discretization error for GBM European, Asian-fixing, and GBM cliquet simulations.

2.3 SLV transition

$$dS_i/S_i = (r - q_i) dt + L_i(S_i) \sqrt{v_i} dW_i^S \\ dv_i = \kappa_i(\theta_i - v_i) dt + \xi_i \sqrt{v_i} dW_i^v.$$

The experiments use full-truncation Euler. With $v_{\text{plus}} = \max(v, 0)$:

$$S_{\text{next}} = S \exp((r - q - 0.5 L(S)^2 v_{\text{plus}}) dt \\ + L(S) \sqrt{v_{\text{plus}}} dt Z_S)$$

$$v_{\text{next}} = \max(v + \kappa(\theta - v_{\text{plus}}) dt \\ + \xi \sqrt{v_{\text{plus}}} dt Z_v, 0).$$

This preserves positivity numerically but introduces discretization bias. Proxy error is measured against a benchmark using the same grid, so Euler bias is outside the reported proxy metric.

3. Monte Carlo labels and variance reduction

3.1 Plain estimator and uncertainty

```
Y_j = discounted payoff on path j
V_hat_MC = (1/N) Sum_j Y_j
SE(V_hat_MC) = sample_std(Y) / sqrt(N).
```

The standard error falls only as $N^{(-1/2)}$. Ten times smaller noise requires one hundred times as many paths, so structural variance reduction is more valuable than path count alone.

3.2 Antithetic sampling

For $Z \sim N(0, I)$, both Z and $-Z$ have the same law.
Use pair average: $Y_{\text{pair}} = (Y(Z) + Y(-Z))/2$.

$\text{Var}(Y_{\text{pair}}) = 0.5 \text{Var}(Y) + 0.5 \text{Cov}(Y(Z), Y(-Z))$.

For monotone payoffs the covariance is often negative, reducing variance without changing the expectation.

3.3 Likelihood-ratio importance sampling

Suppose the standard normal is sampled from $q=N(\mu, I)$ instead of $p=N(0, I)$. The density ratio follows directly by completing the square:

```
p(z) / q(z)
= exp(-z'z/2 + (z-mu)'(z-mu)/2)
= exp(-mu'z + mu'mu/2).
```

$E_p[H(Z)] = E_q[H(Z) p(Z)/q(Z)]$.

For multiple independent steps, log likelihoods add. The European and Asian experiments shift normals toward the payoff boundary. SLV cliquets use a defensive mixture: half unshifted and half shifted paths.

$q_{\text{mix}}(z) = 0.5 p(z) + 0.5 p(z-\mu)$

```
p(z)/q_mix(z)
= 1 / [0.5 + 0.5 exp(mu'z - mu'mu/2)].
```

The unshifted component protects central states; the shifted component stabilizes rare positive payoffs near a global floor.

4. Control variates

Let Y be the target payoff and C a correlated control with known expectation c . Define $Y_{\text{beta}} = Y - \text{beta}(C - c)$. It remains unbiased because $E[C - c] = 0$.

$$\text{Var}(Y_{\text{beta}}) = \text{Var}(Y) + \text{beta}^2 \text{Var}(C) - 2 \text{beta} \text{Cov}(Y, C).$$

Differentiate with respect to beta :

$$2 \text{beta} \text{Var}(C) - 2 \text{Cov}(Y, C) = 0.$$

$$\text{beta}_{\text{star}} = \text{Cov}(Y, C) / \text{Var}(C).$$

The Asian benchmark uses the exact discrete geometric Asian value as the control expectation. The GBM cliquet uses the future clipped-return sum, whose first moment is available from truncated lognormal moments.

4.1 Truncated lognormal moment identity

If $X \sim N(\mu, s^2)$, then for any real a and threshold b :

$$\begin{aligned} E[\exp(aX) \mathbf{1}_{\{X \leq b\}}] \\ &= \exp(a\mu + 0.5 a^2 s^2) \\ &\quad \Phi\left(\frac{b - \mu - a s^2}{s}\right). \end{aligned}$$

Derivation: multiply the normal density by $\exp(aX)$, complete the square, and recognize a shifted normal density. Differences of this expression over two thresholds give interval moments. These moments produce exact means and variances for $\text{clip}(\exp(X) - 1, \text{local_floor}, \text{local_cap})$.

4.2 Benchmark independence

Training and benchmark random streams are independent. A proxy should not be evaluated against the same paths used to train it; common random numbers may make a poor proxy appear artificially accurate.

5. Proxy targets and regularized bases

5.1 Coordinate scaling

$$x_{\text{scaled}} = 2 (x - x_{\min}) / (x_{\max} - x_{\min}) - 1.$$

Chebyshev and Bernstein systems are evaluated on compact intervals. Inputs outside the trained domain are clipped in these experiments; production systems should define explicit extrapolation policies.

5.2 Ridge regression

$$\min_{\beta} ||A \beta - y||_2^2 + \lambda ||\Gamma \beta||_2^2$$

Normal equation:

$$(A'A + \lambda \Gamma' \Gamma) \beta = A'y.$$

The intercept is not penalized. Ridge stabilizes noisy Monte Carlo labels and ill-conditioned high-order bases.

5.3 Chebyshev basis

$$\begin{aligned} T_0(x) &= 1, \quad T_1(x) = x, \\ T_{n+1}(x) &= 2x T_n(x) - T_{n-1}(x). \end{aligned}$$

$$\text{Proxy: } \hat{y}(x) = \sum_{k=0}^d \beta_k T_k(x).$$

Chebyshev polynomials are better conditioned on $[-1, 1]$ than monomials and work well for globally smooth value functions.

5.4 Bernstein and global Bezier basis

$$\begin{aligned} B_{k,n}(u) &= C(n, k) u^k (1-u)^{n-k}, \quad 0 \leq u \leq 1. \\ \hat{y}(u) &= \sum_{k=0}^n c_k B_{k,n}(u). \end{aligned}$$

The Bernstein basis is nonnegative and sums to one. With fitted coefficients c_k it is a global Bezier curve. It provided the best European 1D result.

5.5 Log and bounded-logit targets

$$\text{Positive value: } y = \log(V + \epsilon), \quad \hat{V} = \exp(\hat{y}) - \epsilon.$$

Known bounds $L < U$:

$$\begin{aligned} p &= (V - L) / (U - L), \quad y = \log(p / (1 - p)), \\ \hat{V} &= L + (U - L) / (1 + \exp(-\hat{y})). \end{aligned}$$

The log target prioritizes relative accuracy. The logit target guarantees hard global floor/cap bounds for cliquets.

6. PCHIP, Akima, and piecewise Bezier curves

6.1 Cubic Hermite segment

```
t=(x-x_i)/h, h=x_(i+1)-x_i  
  
H(t)=h00(t)y_i + h10(t)h d_i  
      + h01(t)y_(i+1) + h11(t)h d_(i+1)  
  
h00=2t^3-3t^2+1, h10=t^3-2t^2+t  
h01=-2t^3+3t^2, h11=t^3-t^2.
```

6.2 PCHIP slope rule

Let secants $\delta_i = (y_{i+1} - y_i)/h_i$. If adjacent secants have opposite sign, PCHIP sets the interior derivative to zero. Otherwise it uses a weighted harmonic mean:

```
d_i = (w1+w2)/(w1/delta_(i-1)+w2/delta_i)  
w1=2h_i+h_(i-1), w2=h_i+2h_(i-1).
```

This suppresses overshoot and preserves monotonicity. It is especially useful for American continuation curves near an exercise boundary.

6.3 Akima slope rule

```
d_i = [w_left delta_(i-1) + w_right delta_i]  
      / (w_left+w_right)  
  
w_left = |delta_(i+1)-delta_i|  
w_right = |delta_(i-1)-delta_(i-2)|.
```

Akima weights the less rapidly changing side more heavily. It is local and less aggressively monotone than PCHIP. This matched the reduced Asian curve very well.

6.4 Bezier/Hermite equivalence

```
For the same endpoint slopes:  
P0=y_i  
P1=y_i + h d_i/3  
P2=y_(i+1) - h d_(i+1)/3  
P3=y_(i+1).  
  
B(t)=(1-t)^3 P0 + 3(1-t)^2 t P1  
      +3(1-t)t^2 P2 + t^3 P3.
```

Expanding $B(t)$ gives the Hermite polynomial exactly. Therefore a piecewise Bezier curve built from PCHIP slopes is PCHIP in another representation, not a distinct smoother.

6.5 Expanded 99-case smoother study

Seventeen distinct one-dimensional estimators were tested over European, American, Asian, and barrier parameter combinations. Every fitted surface used a 10M-scenario training budget; MC benchmarks used 500,000 paths per state. Product families received equal aggregate weight.

Method	Balanced p99	Max for $V \geq 0.05$	Local overshoot
Natural cubic interpolation	1.799%	1.163%	2.01%
PCHIP	1.834%	1.220%	0.00%
Akima	1.958%	1.286%	0.17%
MAKIMA	1.968%	1.317%	0.15%
Cubic smoothing spline	2.098%	1.440%	2.49%
Linear interpolation	2.683%	2.058%	0.00%

Natural cubic interpolation led average accuracy by only 0.035 percentage points. PCHIP won 53 of 99 head-to-head p99 cases and introduced no local overshoot, so PCHIP remains the generic one-feature PFE default.

7. European option proxy

7.1 State and benchmark

For constant-parameter GBM, the state at a fixed time is spot alone. The benchmark is the Black-Scholes formula. For a call:

```
d1=[log(S/K)+(r-q+sigma^2/2)tau]/(sigma sqrt(tau))
d2=d1-sigma sqrt(tau)
C=S exp(-q tau) Phi(d1)-K exp(-r tau) Phi(d2).
```

7.2 Why d1 is the fitting coordinate

d1 combines spot, strike, volatility, carry, and remaining maturity into the standardized coordinate controlling exercise probability and delta. Sampling uniformly in d1 is equivalent to deliberate wing coverage in delta space.

7.3 Training label

For each d1 state, shifted antithetic terminal normals produce an unbiased MC label. The default target is $\log(\text{value})$, originally fitted by degree-7 Chebyshev ridge.

7.4 One-dimensional method comparison

Method	Worst max error	Average p99	Average MAE
Global Bernstein/Bezier	0.536%	0.208%	0.003546
Chebyshev	0.854%	0.326%	0.004423
Akima	2.540%	1.127%	0.007015
PCHIP	2.554%	1.149%	0.007098
Bezier with PCHIP slopes	2.554%	1.149%	0.007098

Conclusion: the European curve is sufficiently smooth that global regularized bases average MC noise better than exact local interpolation.

For implementation consistency, the standalone universal default is now log-PCHIP in d1. Its independent-seed worst error is 2.637%, versus the product-specific Bernstein result of 0.536%.

8. Arithmetic Asian option proxy

8.1 Raw Markov state

At fixing index j , store current spot S and running sum A of fixings strictly before today. Let N be total fixings and $m=N-j-1$ future fixings.

$$\text{Payoff} = [(A + S + \text{Sum}_{(k=1)}^m S_{(j+k)})/N - K]_+.$$

8.2 Adjusted-strike derivation

$$\begin{aligned} \text{Payoff} \\ &= (m/N) [(1/m) \text{Sum}_{(k=1)}^m S_{(j+k)} - K_{\text{adj}}]_+ \end{aligned}$$

$$K_{\text{adj}} = (N K - A - S)/m.$$

This is an algebraic identity. It rotates the diagonal payoff boundary in (S,A) into a strike-like scalar.

8.3 GBM homogeneity reduction

Future GBM spots satisfy $S_{(j+k)}=S G_k$, where the joint growth vector G does not depend on S . Therefore:

$$\begin{aligned} V(S,A,j) \\ &= (m/N) E [(S \text{ average}(G) - K_{\text{adj}})_+] \\ &= S (m/N) E [(\text{average}(G) - K_{\text{adj}}/S)_+]. \end{aligned}$$

At a fixed date the normalized value V/S is a one-dimensional function of K_{adj}/S , represented by an adjusted d1-like coordinate. This is why a generic 2D fit is unnecessary for this GBM product.

8.4 Exact linear wing

If $K_{\text{adj}} \leq 0$ for a call, every future arithmetic average is positive relative to the adjusted strike, so the positive-part operator can be removed. The value is then the discounted expected average minus strike and is computed analytically.

9. Asian benchmark and fitting results

9.1 Geometric Asian control

The log of the discrete future geometric average is a weighted sum of normal increments, hence normal. If weights are $w_l = (m-l+1)/m$:

$$\begin{aligned} \log(G_{\text{geo}}) = & \log(S) \\ & + (r-q-\sigma^2/2) \, dt \, (m+1)/2 \\ & + \sigma \sqrt{dt} \sum_l w_l Z_l. \end{aligned}$$

$$\text{Var}(\log(G_{\text{geo}})) = \sigma^2 \, dt \sum_l w_l^2.$$

Thus the geometric Asian call has a Black-Scholes-like exact value and is a highly correlated control for the arithmetic payoff.

9.2 Hybrid target

Fit $\log(V/S)$ in the OTM and moderate region. In the deep ITM region fit $\log((V\text{-linear_baseline})/S)$, then add the exact baseline. This prevents a large intrinsic component from hiding time-value error.

9.3 1D fitter comparison

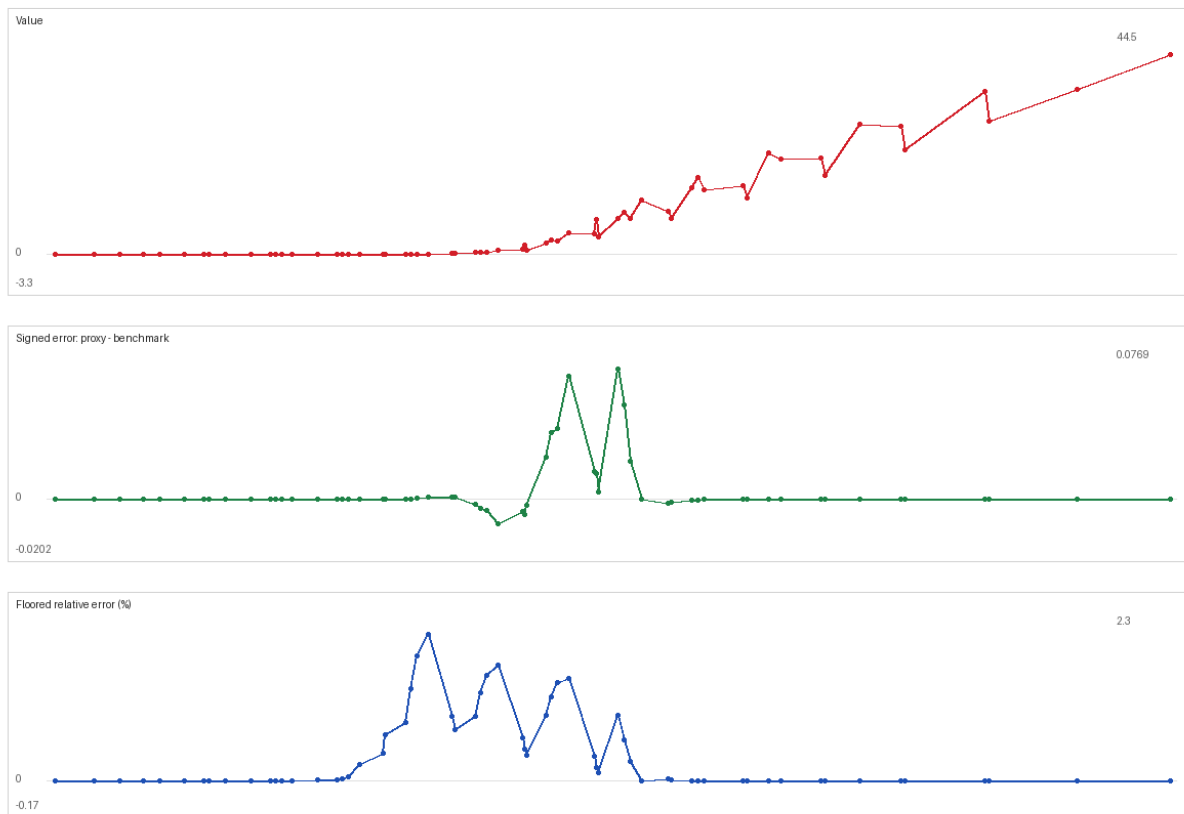
Method	Worst max error	Average p99	Average MAE
Akima	0.186%	0.057%	0.000178
Chebyshev	3.286%	1.721%	0.013523
PCHIP	3.453%	0.936%	0.000273
Global Bernstein/Bezier	6.854%	3.303%	0.017907
Bezier with PCHIP slopes	3.453%	0.936%	0.000273

Akima is the preferred 1D fitter for this reduced Asian experiment. PCHIP is usable, but its monotonicity limiter can flatten local derivatives when MC labels contain small slope reversals. A second independent-seed run confirmed Akima at 0.144% worst error, versus 2.577% for Chebyshev and 1.643% for PCHIP.

The standalone universal default is nevertheless PCHIP. With its independent default seed it achieved 3.331% worst error while sharing the same fitting engine as European and American options.

Asian hybrid proxy validation - day 6, 5 fixings remaining after today
x-axis: adjusted d1 moneyness

black: benchmark, red: proxy



Asian adjusted-moneyness proxy diagnostic from the broader experiment.

10. American put: optimal stopping

10.1 Snell envelope and Bellman recursion

For exercise dates t_k , intrinsic payoff $g(S)=(K-S)_+$. The value is the smallest supermartingale dominating g , equivalently the Snell envelope:

$$\begin{aligned}V_M(S) &= g(S) \\ C_k(S) &= \exp(-r \, dt) \, E[V_{k+1}(S_{\text{next}}) \mid S_k=S] \\ V_k(S) &= \max(g(S), C_k(S)).\end{aligned}$$

The max creates a moving derivative kink at the exercise boundary. A global polynomial fitted recursively can turn a small local continuation error into a systematic bias at every earlier date.

10.2 MC dynamic programming

At each of 100 exercise dates, the implementation uses 121 log-spaced state nodes, antithetic one-step GBM transitions, and about 10 million transitions over the complete backward pass. It fits C_k , then imposes $\max(g, C_k)$ exactly.

10.3 Why PCHIP wins here

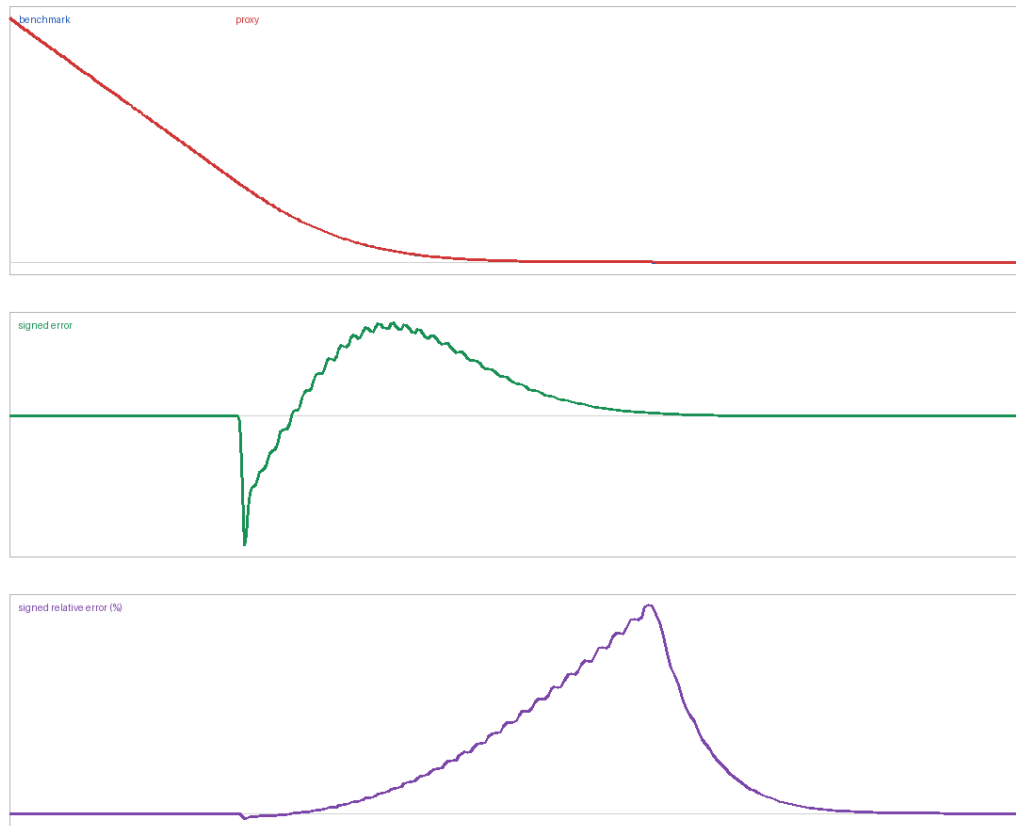
PCHIP is local, shape preserving, and does not ring around the moving boundary. The broad experiment achieved 2.249% worst max error; an independent seed achieved 5.289%. Global log-Chebyshev fits were unstable because exponentiated errors recursively compounded.

Akima was also tested through the complete 100-date recursion. It achieved 3.225% worst max error, 1.675% average p99, and 0.002033 average MAE, versus 2.249%, 0.947%, and 0.001479 for PCHIP. Akima is viable, but PCHIP's monotonicity-preserving slopes are safer near the exercise boundary.

10.4 Independent benchmark

A projected implicit finite-difference solver uses 4,000 time steps and 2,000 spot steps. Projected SOR enforces $V \geq g$. At $S=100$, $t=0$, the finite-difference value is about 6.6597 versus 6.6605 from a separate 4,000-step CRR tree.

American put proxy, time=0.40



American put value, signed error, and signed relative error at time 0.4.

10A. Single and double barrier options

10A.1 State and variants

The experiment prices zero-rebate down-and-out, up-and-out, and double-knock-out calls with either monthly discrete or continuous monitoring. Conditional on no historical hit, spot is the only fitted feature. A live/hit indicator must be added once monitoring history exists.

Zero-rebate knock-ins require no second MC training run. Under the same monitoring convention, in/out parity gives $V_{in} = V_{vanilla} - V_{out}$.

10A.2 Exact Brownian-bridge segment correction

For log endpoints x and y above lower log barrier b over a GBM step with variance $q=\sigma^2 dt$, reflection gives the conditional survival probability

$$P(\min \text{ bridge} > b \mid x, y) \\ = 1 - \exp[-2 (x-b)(y-b)/q], \quad x > b, y > b.$$

The upper-barrier expression replaces the distances by $b-x$ and $b-y$. Multiplying segment survival probabilities and weighting the payoff integrates out crossing indicators, reducing variance. For a double barrier, the absorbing interval transition density is evaluated by a method-of-images series and divided by the free transition density.

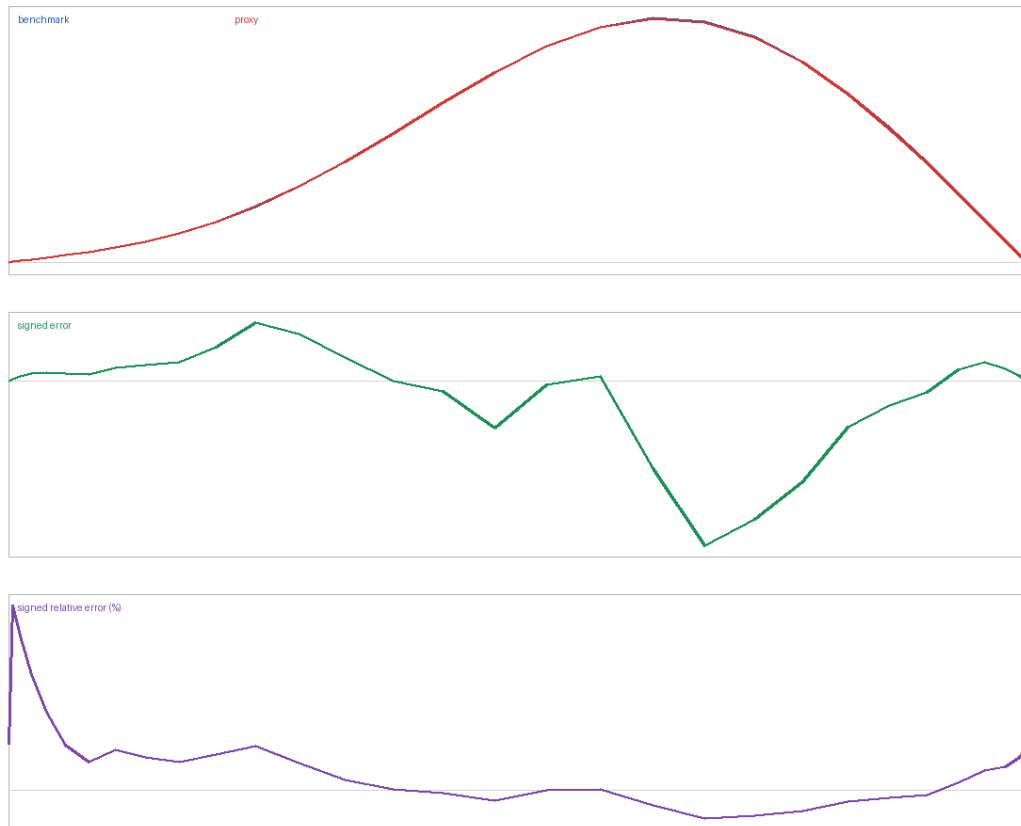
10A.3 Sampling and results

Log-Chebyshev spot nodes cover the alive domain. Common random numbers make label noise smooth across spot, antithetics reduce path noise, and shifted interior validation nodes prevent training-grid leakage.

Variant	PCHIP raw worst	Worst for $V \geq 0.05$
Down-out discrete	2.253%	2.087%
Down-out continuous	2.531%	1.762%
Up-out discrete	9.096%	4.853%
Up-out continuous	8.808%	5.074%
Double-out discrete	8.387%	4.999%
Double-out continuous	8.148%	4.916%

The 8-9% raw maxima are one- or two-cent option values. Meaningful-value errors remain around 2% for down barriers and about 5% for up/double barriers. PCHIP remains the default because the broader 99-case study found near-best accuracy with zero local overshoot.

double_out_continuous reset month 6



Continuous double-knock-out value, signed error, and signed relative error.

11. Single-name GBM cliquet

11.1 Payoff

```
c_i = clip(S_i/S_(i-1)-1, local_floor, local_cap)
H = notional * clip(Sum_i c_i, global_floor, global_cap).
```

11.2 State reduction theorem

Under GBM, each future reset return $S_i/S_{(i-1)}-1$ depends only on the new normal increment and fixed model parameters, not on $S_{(i-1)}$. Future coupons are therefore independent of current spot conditional on reset time.

```
State at reset = accrued = Sum of realized clipped coupons.
V = V(accrued, remaining_periods).
```

This is an exact scale-invariance reduction, not an empirical approximation.

11.3 Expected-total coordinate

```
z = [accrued + m E(c) - midpoint(global bounds)]
    / sqrt(m Var(c)).
```

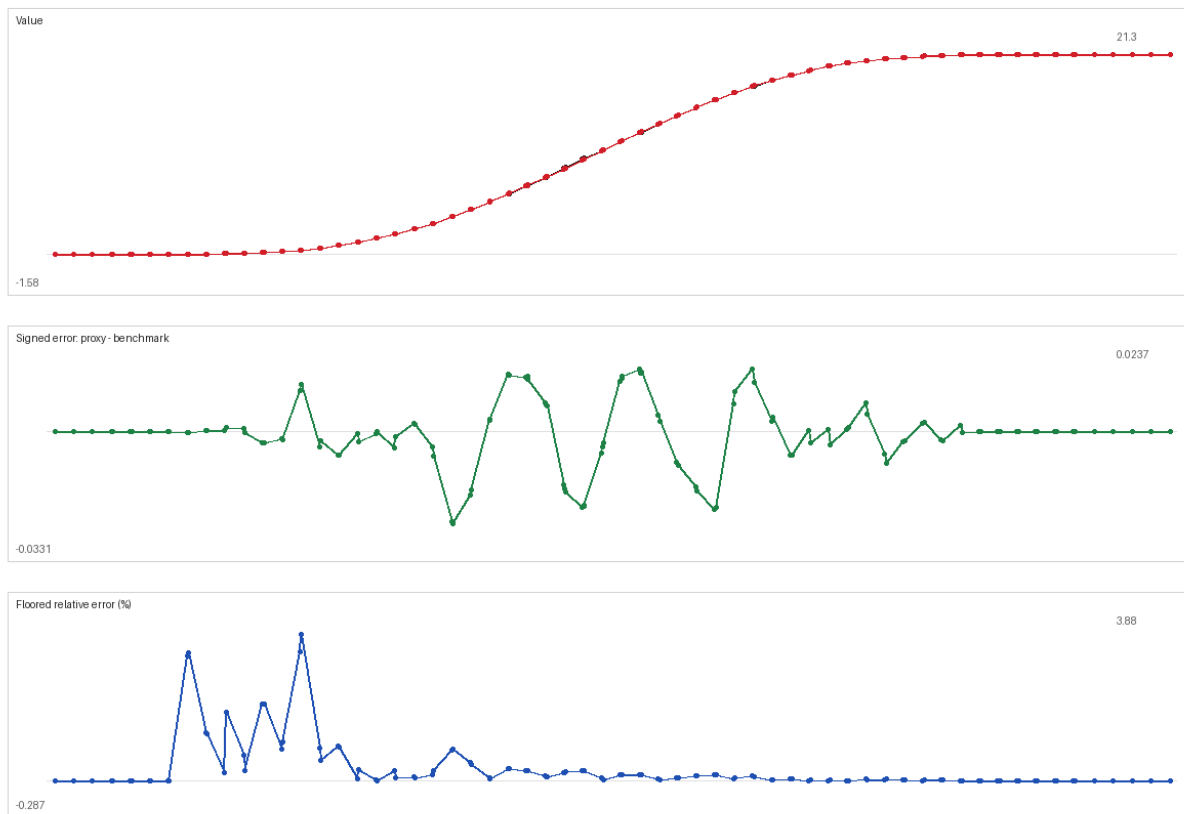
A boundary-enriched grid samples around the estimated floor and cap transitions. Known flat tails are exact: if even all remaining local caps cannot leave the global floor, or all local floors force the cap, regression is bypassed.

11.4 Result

Bounded-logit Chebyshev degree 19 achieved 3.593% worst max error, 0.928% average p99, and 0.003194 average MAE.

Cliquet proxy validation - day 9, 3 periods remaining
method: logit_z_boundary_d19

black: benchmark, red: proxy



GBM cliquet proxy diagnostic with three coupons remaining.

12. Single-name SLV cliquet

12.1 Expanded state

SLV destroys GBM scale invariance because leverage $L(S)$ depends on spot and future variance depends on current v . The reset state becomes (accrued, S , v).

12.2 Structured features

A frozen-coefficient one-period coupon distribution supplies conditional mean and variance. These form an expected-total coordinate z , supplemented by $\log(S/S_0)$ and $\log(v/\theta)$. The target remains bounded logit.

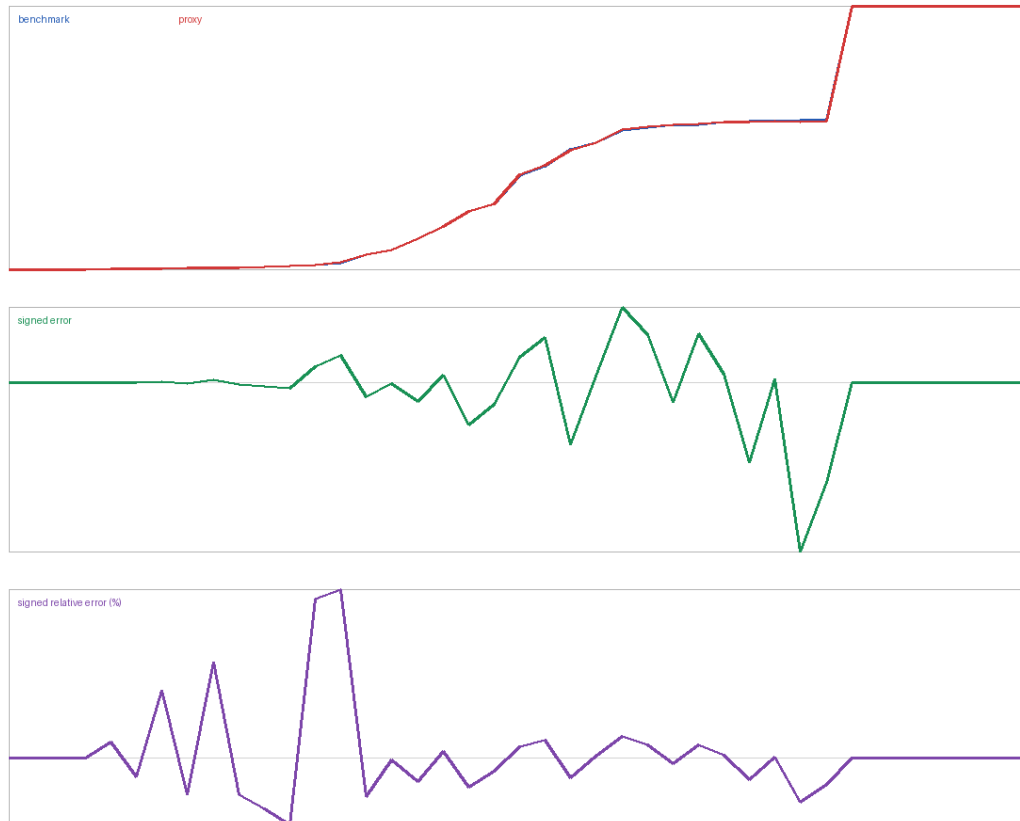
12.3 Maturity-adaptive fit

Remaining coupons	Fitter	Reason
9-12	Local quadratic	Broad smooth long-horizon dependence
3-6	Anisotropic Chebyshev	Sharper payoff transitions
0	Exact payoff	No continuation uncertainty

Lower-tail mixture importance sampling was necessary for seed stability. The common-random-number rerun achieved 5.067% worst error. A separate two-design study rebuilt 10M-path labels and used 500,000 benchmark paths per state on both designs; the fixed adaptive hybrid achieved 4.795% worst error.

Anisotropic Chebyshev degrees 13-23, sparse Hermite, local regression, and Nystrom Matern regression were tested in that study. None beat the existing time-based local/spectral rule. Common random numbers are retained because they smooth label noise across state without changing conditional means.

SLV cliquet, reset month 9, validation states ranked by benchmark



Single-name SLV cliquet validation states ranked by benchmark value.

13. Three-underlying SLV basket cliquet

13.1 Three coupon definitions

Variant	Monthly coupon
Basket return	$\text{clip}(\text{mean}(R1, R2, R3), \text{local floor}, \text{local cap})$
Average clipped	$\text{mean}(\text{clip}(R1), \text{clip}(R2), \text{clip}(R3))$
Worst of	$\text{clip}(\min(R1, R2, R3), \text{local floor}, \text{local cap})$

13.2 Seven-dimensional Markov state

$$X = (\text{accrued}, S1, S2, S3, v1, v2, v3).$$

Each asset has its own variance factor and leverage function. Correlated market drivers create cross-asset dependence; each spot shock also contains its own negatively correlated variance shock.

$$Z_i^S = \rho_i Z_i^V + \sqrt{1 - \rho_i^2} Z_i^M, \\ \text{Corr}(Z^M) = R_{\text{market}}.$$

This construction is automatically positive semidefinite because it is a linear transformation of independent variance normals and a valid correlated market-normal vector.

13.3 Sampling

2,001 low-discrepancy boundary-enriched states cover accrued return, three log spots, and three log variances. Approximately 10 million scenarios are spread over each reset-date fit. Each of 31 independent validation states uses 500,000 paths.

13.4 Features and bounded target

Conditional coupon moments define separate lower and upper standardized cushions. Keeping both is important: one midpoint z cannot distinguish states with equal standardized midpoint but different dispersion and distances to the two global bounds.

14. Basket high-dimensional model search

The search compared local full-state and summary regressions, sparse anisotropic Chebyshev terms, Gaussian RBF kernels, and a two-layer tanh neural ensemble. With averaged state labels, neither RBF nor the small neural ensemble was uniformly competitive. This does not reject neural networks in general; it shows that architecture and state coverage matter.

14.1 Development ensemble and validation failure

```
V_proxy = w(m) V_local + [1-w(m)] V_spectral  
w(m) = 0.16 + 0.02 m,  
m = remaining coupons.
```

For symmetric basket-return and average-clipped coupons, V_{local} used summary features. Worst-of used the full state. The blend performed well on the first development design but did not generalize, so it was retired.

Variant	Development	Validation 2	Validation 3	Worst
Basket return	10.258%	7.108%	18.500%	18.500%
Average clipped	5.624%	11.566%	5.745%	11.566%
Worst of	9.700%	9.862%	6.774%	9.862%

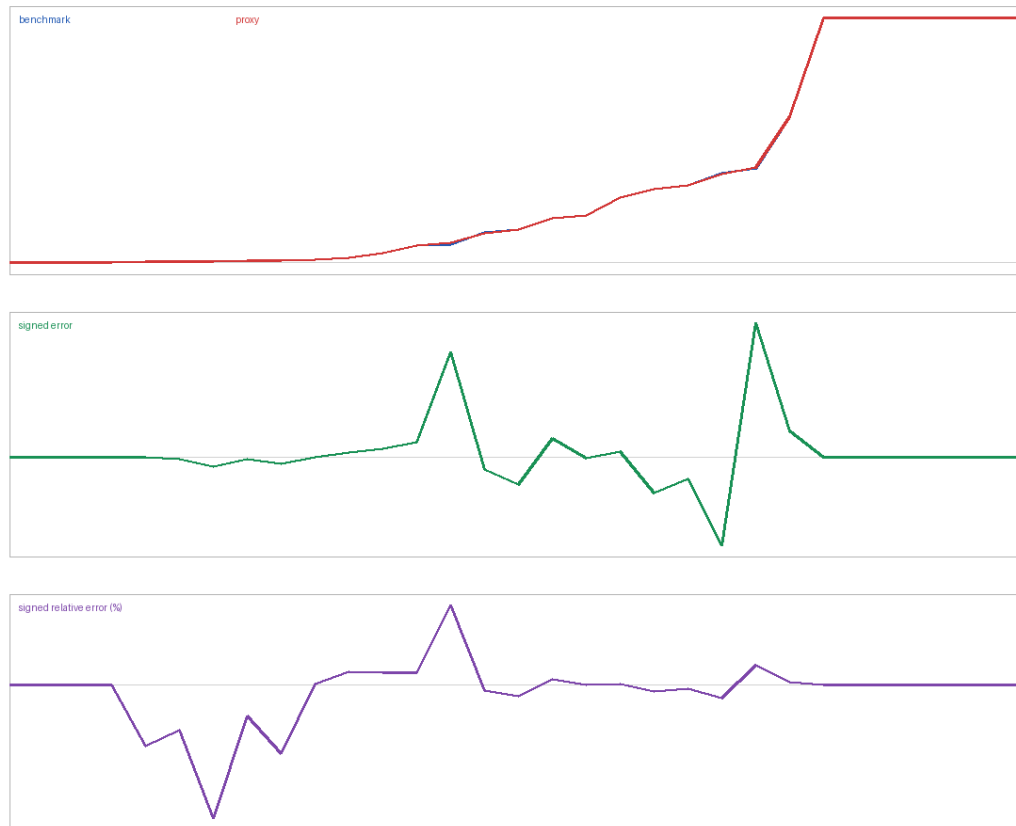
These are fixed sparse-spectral results using 2,001 states. A 5,001-state design, local models, RBF kernels, small tanh networks, random-feature networks, and bagging were also tested. None guaranteed 5-8% on the third untouched design. The next credible step is adaptive state enrichment around failed neighborhoods, followed by another untouched validation design.

14.2 Literature-inspired residual search

A clipped-normal moment baseline was used as a low-fidelity model, followed by direct or residual sparse Hermite, local, Nystrom Matern, and fixed ensemble corrections. Coupon skewness and floor/cap masses were added as features. Inverse-variance weighted sparse Chebyshev fits were also tested on both 2,001 and 5,001 states.

None improved worst-case relative error on the development and independent designs. Some dense weighted fits reduced MAE but leaked larger errors into tail states. The moment baseline did not simplify correlated clipped and worst-of SLV coupons enough for residual learning to help.

3-asset SLV basket, return, reset month 9, states ranked by benchmark



Development diagnostic for the fixed sparse basket-return proxy with three coupons remaining.

15. Generic workflow for a new instrument

Step 1: define the contract exactly

- Observation schedule, exercise rights, local/global bounds, and settlement.
- Clarify basket aggregation order: clip-then-average is not average-then-clip.

Step 2: identify a sufficient Markov state

- Start from the simulator state and add path statistics required by payoff.
- Prove any dimension reduction from scale invariance or payoff algebra.

Step 3: nondimensionalize

- Use moneyness, normalized variance, expected-total cushions, and time.
- Align transition boundaries across maturities before fitting.

Step 4: exploit exact regions

- Terminal payoff, linear ITM wings, hard global floor/cap tails, intrinsic value.

Step 5: design state sampling

- Combine broad low-discrepancy coverage with boundary-focused clusters.
- Do not spend most paths in flat regions that are already known analytically.

Step 6: generate MC labels

- Use antithetic paths by default.
- Add controls when exact expectations are available.
- Use likelihood-ratio shifts for rare payoff wings.

Step 7: choose fitter by effective dimension

Effective dimension	Default candidates
Any genuine 1D price proxy	PCHIP operational default
1D smooth, product optimized	Natural cubic, Chebyshev, Bernstein
1D noisy labels	Smoothing spline or P-spline after validation
2-5D smooth	Sparse anisotropic basis, local polynomial, small NN
6D+	Smooth NN or sparse/additive model; ensemble after validation

Step 8: validate independently

- Use new random seeds and a stronger benchmark.
- Report value, signed error, relative error, and benchmark standard error.
- Slice results by maturity, payoff region, and state-space boundary.
- For PFE, run the proxy through outer scenarios and compare exposure quantiles by date.

16. Why there is no universal fitter

PCHIP works well for the American continuation curve because the geometry is one-dimensional, monotone, and kinked. It is not directly defined on a general multidimensional cloud. Tensor-product PCHIP would grow exponentially with dimension and requires a structured grid.

Chebyshev and Bernstein bases are excellent global smoothers in one dimension. In seven dimensions, unrestricted tensor bases are combinatorial. Sparse term selection helps, but sharp payoff transitions and interactions can still produce bias.

Neural networks are a genuinely generic high-dimensional candidate because parameter count can grow without a tensor grid. Their success requires enough state diversity, regularization, architecture tuning, and independent seeds. The small basket network tested here did not beat the structured ensemble.

A robust universal *workflow* is more realistic than a universal fitter: state reduction, payoff-aware features, bounded targets, variance-reduced labels, dimension-appropriate smoothers, and independent validation.

Recommended hierarchy

Question	Action
Can the state be proved 1D?	Use and compare global ridge, Akima, PCHIP.
Are hard bounds known?	Use exact tails and bounded-logit target.
Is there a dominant baseline?	Fit residual or time value.
Do local/global models err oppositely?	Use a simple cross-validated blend.
Does dimension exceed sparse-basis capacity?	Increase state coverage and test smooth NN.

17. Error accounting

17.1 Metrics

```
signed_error = proxy - benchmark
absolute_error = |signed_error|
relative_error = absolute_error / max(|benchmark|, 0.01)
noise_ratio = absolute_error / max(benchmark_SE, tiny).
```

A low noise ratio indicates the discrepancy may be statistically unresolved. A large noise ratio indicates model/fitting bias rather than benchmark noise.

17.2 Error decomposition

```
proxy - true continuous-model value
= fitting error
+ training MC error
+ benchmark estimation comparison noise
+ time-discretization bias
+ model/calibration error.
```

The reported studies isolate fitting plus training noise against a benchmark under the same model discretization. They do not claim market calibration accuracy.

17.3 Leakage controls

- Independent training and benchmark random streams.
- Fixed validation states not reused as training states.
- A final standalone run with an independent seed where provided.
- Method-selection results should be confirmed on a second validation design.

17.4 Stability checks

- Repeat labels with different seeds.
- Double Euler steps for SLV and compare.
- Expand spot/variance domains to test extrapolation.
- Check price bounds and monotonicity numerically.
- For Greeks, validate derivative smoothness separately from price error.

18. Reproducibility map

Purpose	Entry point or folder
European default	EuroMain.py
Asian default	AsianMain.py
Barrier research/default	BarrierOptExperiment/BarrierMain.py
GBM cliquet default	CliquetMain.py
American put default	AmericanMain.py
Single-name SLV cliquet	SLVCliquetMain.py
3-asset SLV basket cliquet	BasketCliquetOptExperiment/BasketCliquetMain.py
Expanded 1D comparison	OneDimensionalFitExperiment/ExpandedSplineStudy.py
European research	EuroOptExperiment/
Asian research	AsianOptExperiment/
Cliquet research	CliquetOptExperiment/
American research	AmericanOptExperiment/
SLV research	SLVCliquetOptExperiment/ and BasketCliquetOptExperiment/

Core numerical budgets

Study	Training	Benchmark
European	121 states x 25,000 shifted paths	Black-Scholes closed form
Asian	about 10M scenarios per fitted date	500K paths/state + geometric control
GBM cliquet	about 10M scenarios per fitted date	500K paths/state + clipped-sum control
American	10,006,700 one-step transitions	4,000 x 2,000 projected FD grid
Barrier	10M scenarios per fitted date	500K paths/state + bridge survival
Single SLV cliquet	about 10M scenarios per fitted date	500K paths/state
Basket SLV cliquet	about 10M scenarios over 2,001 states/date	500K paths x 31 states/date

Known limitations

- Illustrative SLV leverage functions are not calibrated market surfaces.
- SLV uses two Euler steps per monthly period in the current experiments.
- Basket 5-8% control failed on untouched designs; adaptive enrichment remains research.
- Pointwise price accuracy does not replace outer-scenario PFE quantile validation.
- Production extrapolation, Greeks, and calibration loops require separate tests.
- The neural basket test is not a definitive neural-network architecture search.

19. Final conclusions

The strongest improvement repeatedly came from changing the representation before changing the regressor. d1 organizes European wings; adjusted strike reduces the GBM Asian state; expected-total cushions organize cliquet bounds; the Snell recursion isolates American continuation.

The expanded one-dimensional test covered 99 European, American, Asian, and barrier cases with 17 estimators. Natural cubic interpolation led balanced p99 by only 0.035 percentage points, while PCHIP won 53 of 99 head-to-head cases and had zero local overshoot.

When one implementation must cover every genuinely one-dimensional option, PCHIP is therefore the universal default. Product-specific optimized profiles remain available when maximum pointwise accuracy matters. For PFE, the final acceptance metric must be exposure-quantile distortion in actual outer scenarios, not price RMSE alone.

For higher-dimensional SLV baskets, sparse payoff-aware regression was the strongest generic baseline, but three independent state designs showed that the 10M-path budget did not guarantee the 5-8% target. Robust worst errors were approximately 18.5%, 11.6%, and 9.9%.

Moment-residual Hermite, local, Nystrom Matern, weighted spectral, and fixed ensemble alternatives did not improve the 7D basket tail. This supports adaptive state enrichment or a larger path-level neural program rather than further tuning on the same sparse labels.

The reusable methodology is therefore: derive the state, reduce dimension when mathematically valid, enforce exact structure, create unbiased low-noise labels, fit with a dimension-appropriate smoother, and validate independently in the wings and transition regions.

End of methodology guide.